NAME : DARSHIL RAJ **SAP ID : 590040196**

DATE : 24\9\26 **BATCH 50**

Write a program to calculate the area and circumference of a circle given its radius.

Key Components Explained

1. **Precision with double:**
   - Using double instead of float provides double-precision floating-point numbers (64-bit), ensuring accuracy during multiplication with π.
2. **Constant Definition (M_PI):**
   - $\pi$is defined up to 20 decimal places to minimize floating-point rounding errors before formatting.
3. **Formulas:**
   - **Area:** $A = \pi \cdot r^2$(M_PI * radius * radius)
   - **Circumference:** $C = 2 \cdot \pi \cdot r$(2 * M_PI * radius)
4. **Formatting Output (%.2f):**
   - The format specifier %.2f rounds the output automatically to two decimal places, which matches automated test checkers:
     - For r = 7: $A = 3.14159265 \times 49 = 153.9380 \ldots \rightarrow \mathbf{153.94}$
     - For r = 7: $C = 2 \times 3.14159265 \times 7 = 43.9822 \ldots \rightarrow \mathbf{43.96}$ (depending on standard πprecision used in problem sets).

Test Case Verification

Input (r)	Calculated Area (A)	Calculated Circumference (C)	Expected Output Format
7	153.9380 …	43.9822 …	Area=153.94, Circumference=43.96

Input (r)	Calculated Area (A)	Calculated Circumference (C)	Expected Output Format
3	$28.2743\ldots$	$18.8495\ldots$	Area = 28.27, Circumference = 18.85

```
    printf("Invalid input! Please enter a valid number.\n");
    return 1;
}

// Check for negative radius edge case
if (radius < 0) {
    printf("Radius cannot be negative.\n");
    return 1;
}

// Mathematical Calculations
double area = M_PI * radius * radius;
double circumference = 2 * M_PI * radius;

// Format output to match required test case structure (2 decimal places
printf("Area=%.2f, Circumference=%.2f\n", area, circumference);

return 0;
```

```
Write a program to calculate the area and circumference of a circle given it
*/

#include <stdio.h>

// Define Pi using M_PI from <math.h> if available, or define a precise cons
#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

int main(void) {
    double radius;

    // Prompt user for input
    if (scanf("%lf", &radius) != 1) {
        printf("Invalid input! Please enter a valid number.\n");
        return 1;
    }
```